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Intermediate Pack - Improve your Alpha [2/2]

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Use Different Operators

Remember the cross-sectional operators and time-series operators from the [Starter pack](#)? You can use them here. Secondly, you can try out different data fields. We recommend exploring the price volume dataset, model dataset and fundamental dataset. Lastly, you can tinker with different simulation settings. These tips should help improve the performance of your second Alpha.

Divide (/)

We can divide data fields with other data fields.

Imagine market data being a matrix, with each row representing one date and each column representing one stock. For example, the matrix for close price data of stocks in universe US TOP3000 would look like this:

Dates	Close Price of A	Close Price of B	Close Price of C	...
2010-01-04	30.95	25.46	214.01	...
2010-01-05	30.96	25.65	214.38	...
2010-01-06	30.77	25.59	210.97	...
...	...	...	...	...

And the matrix for open data of above stocks would look like this:

	Simulate	Alphas	Learn	Data	Competitive (2)
2010-01-04	30.74	25.78		213.82	...
2010-01-05	30.99	25.52		214.10	...
2010-01-06	30.83	25.64		211.35	...
...	...	...		...	...

Say you enter an Alpha expression like ***close/open*** in the Simulate page found in the Alphas dropdown tab. When you click Simulate, BRAIN will evaluate the Alpha expression against the matrix of market data for each date and each stock.

Rank(x)

Description: the Rank operator ranks the value of the input data x for the given stock among all instruments, and returns float numbers equally distributed between 0.0 and 1.0

Alpha expression: ***rank(sales/assets)***. If company B has a higher asset turnover ratio (sales/assets) than company A, stock B may outperform stock A. The rank operator helps to limit the extreme values of that ratio.

For example:

Stock	A	B	C	D	E
Alpha Expression <i>sales/assets</i>	1	2	3	20	100

The numbers imply that if you have \$126, you must use \$100 to go long stock E (~80% of your total capital). So, your strategy would depend crucially on how the last stock performs. But, isn't that too risky? Applying the rank function to the alpha expression ***rank(sales/assets)***, you get:

Stock	A	B	C	D	E
Alpha Expression <i>rank(sales/assets)</i>	0	0.25	0.5	0.75	1

This time you see that the stock with the largest weight occupies only 40% of your portfolio.

Ts_rank & Ts_delta Operator

Description: ranks the values of x for each instrument over the past d days, then return the rank of the current value. The rank value is equally distributed between 0.0 and 1.0	Description: Returns the difference of x and the x value d days ago
Alpha expression: <i>ts_rank(sales/assets,60)</i> . If company A has a higher asset turnover ratio (sales/assets) compared to its historical data, the company may be growing and its stock price might follow. The <i>ts_rank</i> operator compares the historical value of the ratio over the past 60 days and assigns 1 if it is at its highest, and 0 if it is at its lowest.	Alpha expression: <i>-ts_delta(close,2)</i> . If the stock price of a company has increased over the last 2 days, it may decrease in the future. The <i>ts_delta</i> operator takes the difference of the close price today and the close price 2 days ago.

Visual Illustration of Ts_rank Operator:

Ts_rank Operator

Day 8:

Stock A	Day 5	Day 6	Day 7	Today
Alpha Expression <i>sales/assets</i>	1	2	5	4



Stock A	Day 5	Day 6	Day 7	Today
Alpha Expression <i>ts_rank(sales/assets, 4)</i>	0	0.33	1	0.66

Day 9:

Stock A	Day 6	Day 7	Day 8	Today
Alpha Expression <i>sales/assets</i>	2	5	4	20



Stock A	Day 6	Day 7	Day 8	Today
Alpha Expression <i>ts_rank(sales/assets, 4)</i>	0	0.66	0.33	1

Change Simulation Settings

In your first Alpha simulation, you left the simulation settings on default. Changing certain

LANGUAGE: Fast Expression

INSTRUMENT TYPE: Equity

REGION: USA

UNIVERSE: TOP3000

DELAY: 1

NEUTRALIZATION: Subindustry

DECAY: 4

TRUNCATION: 0.08

PASTEURIZATION: On

UNIT HANDLING: Verify

NAN HANDLING: Off

Save as Default

Apply

Region

Region refers to the market in which the Alpha will simulate trades, for example, the U.S. equity market or Chinese equity market.

Universe

Universe is a set of trading instruments ranked by their liquidity. For example, "US: TOP3000" represents the top 3,000 most liquid stocks in the U.S. market.

Decay

Decay is used for averaging the Alpha signal within a specified time window. The settings perform linear decay on the Alpha. Tip: Decay can be used to reduce turnover, but decay values that are too large will attenuate the signal.

$$\text{Decay_linear}(x, n) = \frac{x[\text{date}] * n + x[\text{date} - 1] * (n - 1) + \dots + x[\text{date} - n + 1]}{n + (n - 1) + \dots + 1}$$

Truncation

Truncation sets the maximum weight for each stock in the overall portfolio. It aims to guard against excessive exposure to movements in individual stocks. The recommended

Market risks and industry specific risks are prevalent risks within equities. However, these risks can be reduced by creating long-short neutral portfolios using a concept called neutralization. After neutralizing the portfolios to market or industry specific groups, no net position is taken with respect to that group, i.e. allotting the same amount of dollars in long (buying) and short (selling) positions. That way, you are less exposed to risk, whether the entire market goes up or down.

When Neutralization = "Market/Industry/Sub-industry" it does the following operation:
 $Alpha = Alpha - mean(Alpha)$ where Alpha is the vector of weights.

Before Neutralization		After Neutralization	
alpha_A	alpha_B	new_alpha_A	new_alpha_B
2	4	-1	1

If the hypothetical book size is 20 million, we would end up investing \$10 million in long positions and \$10 million in short positions. Thus, no net position is taken with respect to the market. In other words, the long exposure cancels out the short exposure completely, making this hypothetical strategy market neutral.

The three different neutralization methods determine which groups are used for neutralizing Alpha values. The correct choice of neutralization depends on the logic or formula used by the Alpha. The results should indicate which neutralization will be most effective.

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